AXIAL PISTON PUMPS / PRODUCT INFORMATION

P180 Axial Piston Pump

Plan the high-flow installation around inlet supply, drive power and service access. Use the published curves at matching speed and pressure; separate maximum flow and pressure values do not establish one simultaneous operating point.

The P180 is a different class of pump from everything below it. It is supplied exclusively in the boosted configuration with a mechanical seal, uses a 13-tooth spline drive shaft instead of a keyed connection, and employs 2" (50.8 mm) SAE Code 61 4-bolt flanged pressure ports. These are not incremental changes; they reflect the engineering demands of sustained 430 L/min (114 US gpm) oil-free water hydraulic flow at 160 bar (2,321 psi).



WaterQuest P180 axial piston water hydraulic pump, 316 stainless steel, 252mm body

At a glance

Parameter / value	Parameter / value
Displacements 135/180/225 cc/rev (13.730 in³/rev)	Max Pressure 160 bar (2,321 psi) continuous
Max Flow 430 L/min (113.59 US gpm)	Max Input Power 114 kW (152.88 hp)
Body OD 252 mm ()	Pressure Ports 2" (50.8 mm) SAE 3000 PSI (206.8 bar (2,999 psi)) 4-bolt flange

Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

Technical data

Specifications and selection data

Parameter	135 cc/rev (8.24 in³/rev)	180 cc/rev (11 in³/rev)	225 cc/rev (13.7 in³/rev)	Note
Displacement	135 cc/rev (8.24 in³/rev)	180 cc/rev (11 in³/rev)	225 cc/rev (13.7 in³/rev)	Fixed
Configuration	Boosted PB only - no unboosted option			Mechanical seal standard
Max Continuous Pressure	160 bar (2,321 psi)			
Max Flow (2000 rpm boosted)	~259 L/min (68.4 US gpm)	~344 L/min (90.9 US gpm)	430 L/min (114 US gpm)	PB configuration
Flow at 1750 rpm, 20 bar (290 psi)	~231 L/min (61 US gpm)	~308 L/min (81.4 US gpm)	~385 L/min (102 US gpm)	
Flow at 1500 rpm, 160 bar (2,321 psi)	~180 L/min (47.6 US gpm)	~264 L/min (69.7 US gpm)	~322 L/min (85.1 US gpm)	
Max RPM	2000 rpm (boosted)			Min 2.5 bar (36.3 psi) inlet required
Min Inlet Supply Pressure	2.5 bar (36.3 psi) at all times			Mandatory for all P180 orders
Max Case Drain (T Port) Pressure	2.5 bar (36.3 psi) maximum			
Max Input Power	~70 kW (93.9 hp)	~93 kW (125 hp)	~114 kW (153 hp)	At 1750 rpm, 160 bar (2,321 psi)
Max Torque at 160 bar (2,321 psi) (1500 rpm)	~375 Nm (277 lbf·ft)	~500 Nm (369 lbf·ft)	~605 Nm (446 lbf·ft)	
Peak Efficiency (225cc (13.7 in³), 1500 rpm)	~90% at 80 to 120 bar (1,160 to 1,740 psi)			Highest in the range
Seal	Mechanical seal standard - no lip seal option			
Drive	13-tooth spline, 8/16 pitch, fillet root, side fit			Not keyed
Max Fluid Temperature	50°C (consult WQS above 50°C (122 °F))			
Min Fluid Temperature	2°C (35.6 °F) (antifreeze option available)			
Housing Material	316 Stainless Steel			
Pressure Ports	2" (50.8 mm) SAE Code 61 4-bolt flange			Not BSPP
T Port (Case Drain)	1" BSPP			To low-pressure sump
Flange / Spigot OD	152.35 / 152.30 mm (6 / 6 in)			
Body Outer Diameter	252 mm (9.92 in) (9")			
Overall Length	389.5 mm (88.5 mm (3.48 in) shaft + 301 mm (11.9 in) body)			

Parameter	135 cc/rev (8.24 in³/rev)	180 cc/rev (11 in³/rev)	225 cc/rev (13.7 in³/rev)	Note
Working Shaft Length	54 mm (2.13 in)			
Shaft Projection	14.50 mm (0.571 in) inner / 30.50 mm (1.2 in) base			
Port Face	193 mm (7.6 in) square, M12 x 18 mm (0.709 in) bolt pattern			42.80 mm (1.69 in) spacing, 77.80 mm (3.06 in) offset
Lifting Point	M12 tapped hole on port face for lifting eye			Always use lifting eye when handling
Inlet Filtration Required	10 µm (394 µin) nominal, 25 µm (984 µin) absolute, beta-10 = 75			
Certifications	Confirm applicability and supplied documentation: ATEX II 2GD ck T6, ISO 9001 FM 87247, F4N			

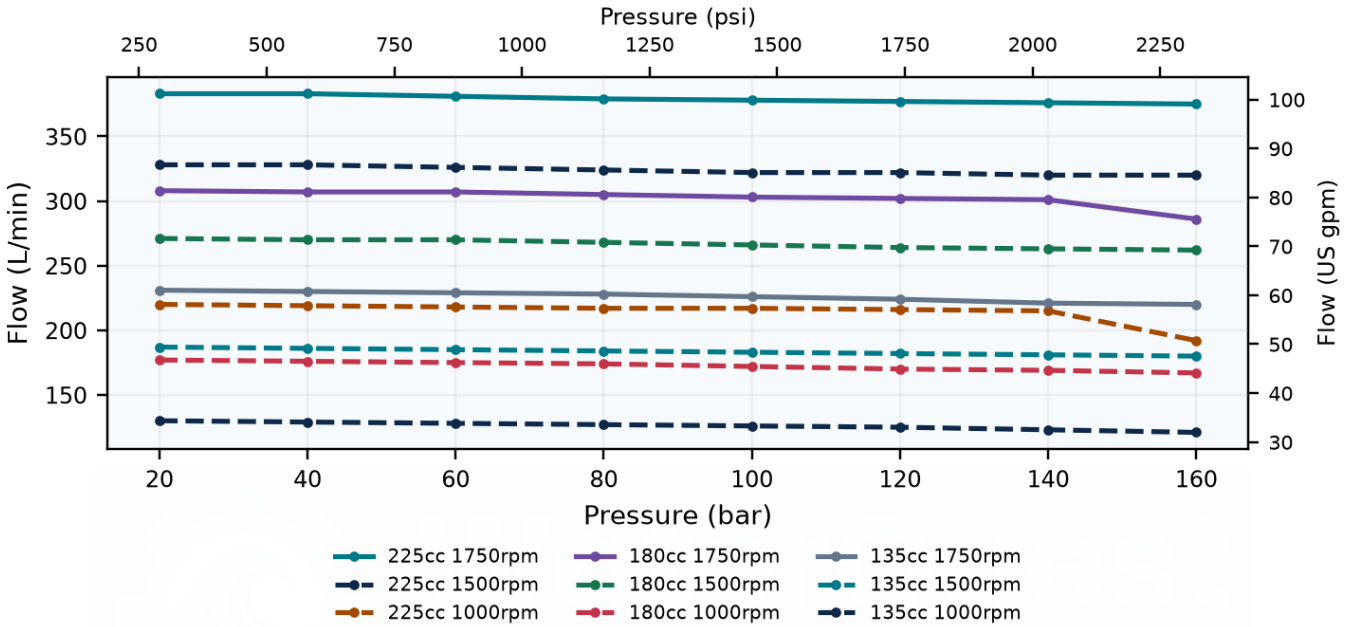
Selection and integration

All performance figures on clean tap water at 15°C (59 °F). The exact operating fluid must be specified on all enquiries and orders without exception. 316 SS Material Test Reports and full dimensional drawings available on request. Contact WaterQuest for motor frame sizing, drive coupling specification, and SAE flange manifold design support.

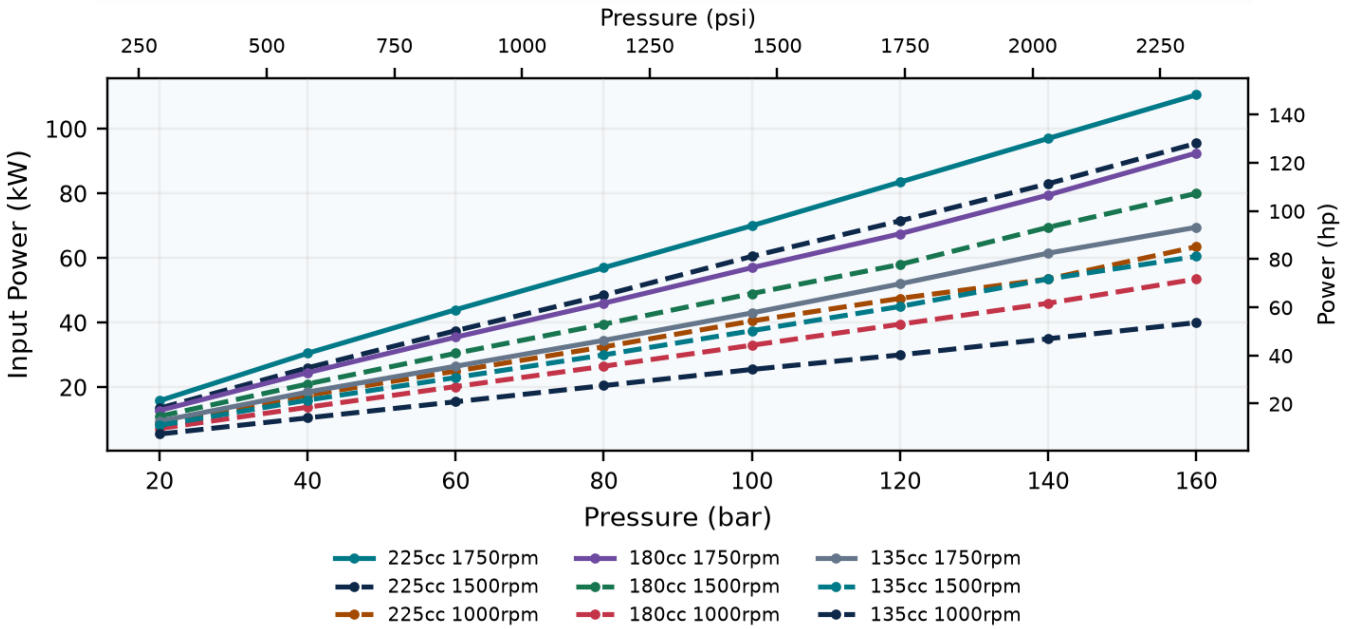
Performance curves

Reference curves reproduced from the product webpage. Curves retain the published data points and speed/displacement labels. Confirm guaranteed performance at the selected duty.

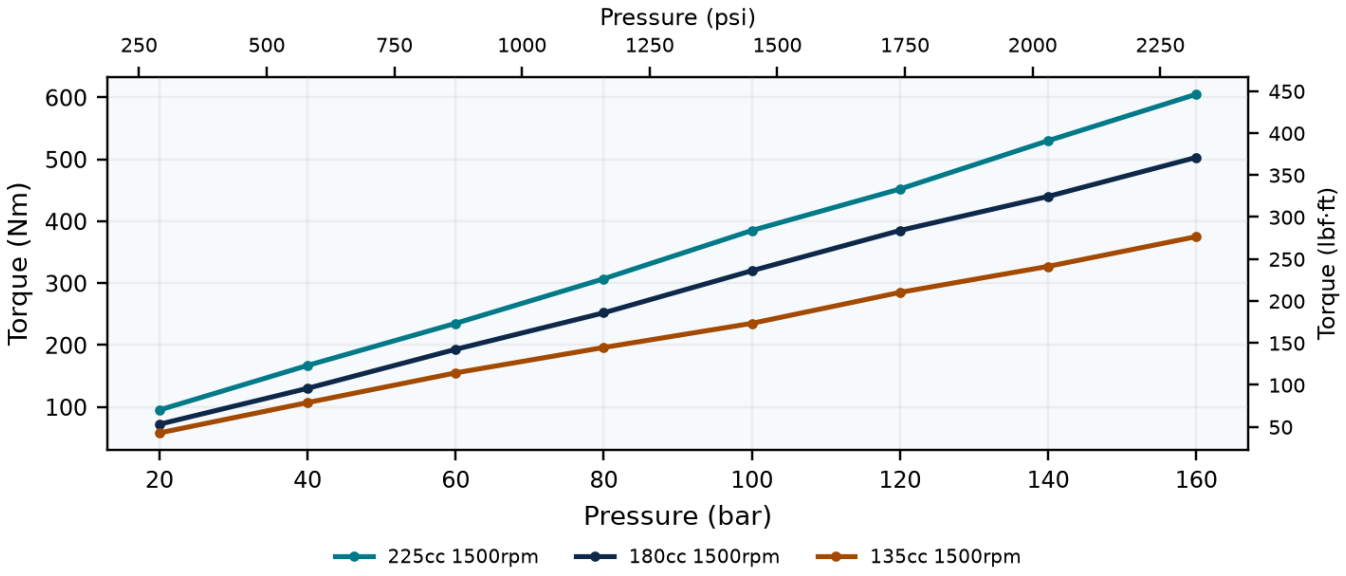
Flow (L/min) vs Pressure (bar)



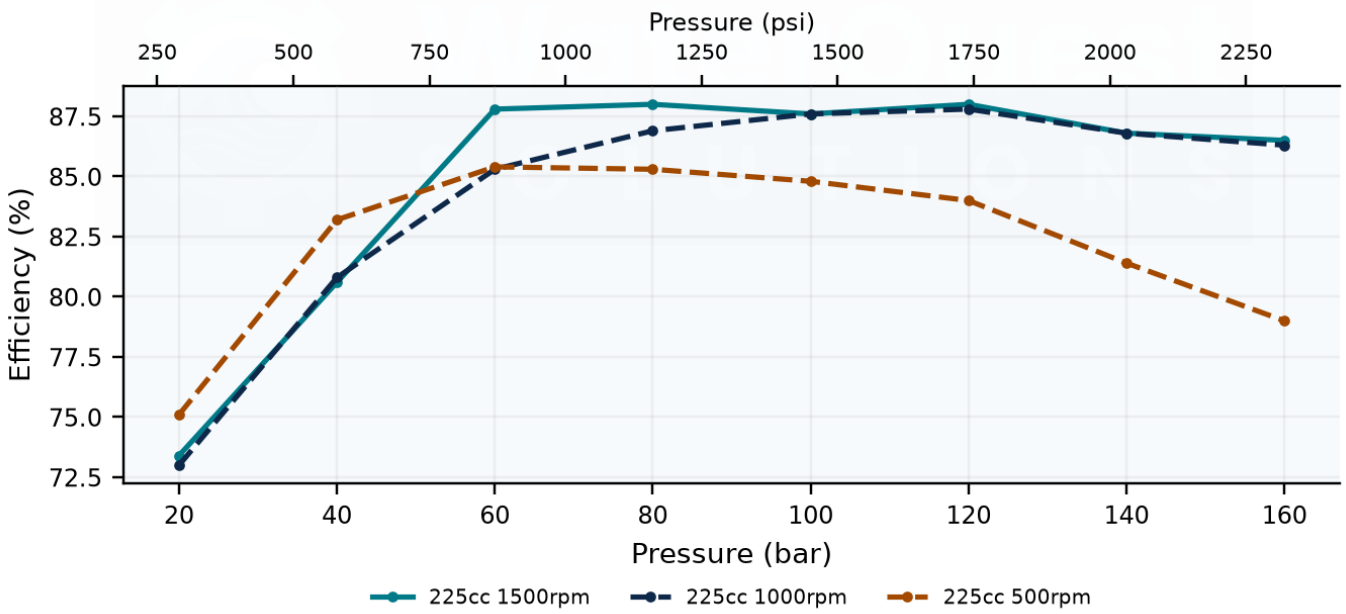
Input Power (kW) vs Pressure (bar)



Torque (Nm) vs Pressure (bar)



Efficiency (%) vs Pressure (bar)



Drawings and reference visuals

Dimensions, ports and curves are reproduced from the webpage visuals. Use the original drawing units and confirm the current approved drawing before fabrication or installation.

PB160-135W, PB160-180W and PB160-225W P180 dimensional drawing, 2 inch SAE 4-bolt flange pressure ports, 1 inch BSPP T port, 13-tooth spline shaft, 252mm body

